

Two Sigma Algorithmic Contributor Program: Market Impact information

The following variables, formula definition, and coefficient are a summary of the Almgren paper.

Variables

I = Permanent Impact = percent change in price between pre and post trade

J = Realized Impact = percent change in price between the realized price and pre trade price (this is the key market impact cost factor that the model generates).

X = Total executed size in shares

T = Trade Duration = in days, measured as a value between 0 and 1

V = Average daily volume in shares

σ = Price Volatility (%)

F = Fill Price for the order (we do not have partial fills)

Θ = Shares Outstanding

Formula/Coefficient Summary

Market Impact = $0.5 * (J * X * F)$

Where I, J are defined as:

$$I = \gamma \sigma \frac{X}{V} \left(\frac{\Theta}{V} \right)^{1/4} + \langle \text{noise} \rangle \quad J = \frac{I}{2} + \text{sgn}(X) \eta \sigma \left| \frac{X}{VT} \right|^{3/5} + \langle \text{noise} \rangle$$

With coefficients

$\gamma = 0.314$

$\eta = 0.142$

$T = 0.0075$

With J capped to 0.1

Algorithmic Contributor Program Deviation From Reference Paper

- Our Algorithmic Contributor Program implementation calculates ADV using a maximum 80-day moving average; the paper uses a 10-day moving average
- We calculate price volatility using daily close prices, the paper uses intraday prices.
 - o This is a simplification to the implementation. The consequence is that our market impact estimate will evolve more slowly than if we used the intraday prices.
- Our implementation disregards noise in the calculation of I and J.
 - o The noise term clarifies how the model can be used to describe/explain observed trade prices, but in our case, where we are *simulating* trade prices, we just want to take the predicted average, and thus we drop the noise term.
- We assume a $T = 0.0075$
 - o We simulate orders going through a mock broker that on average will take 3 minutes to execute an order.